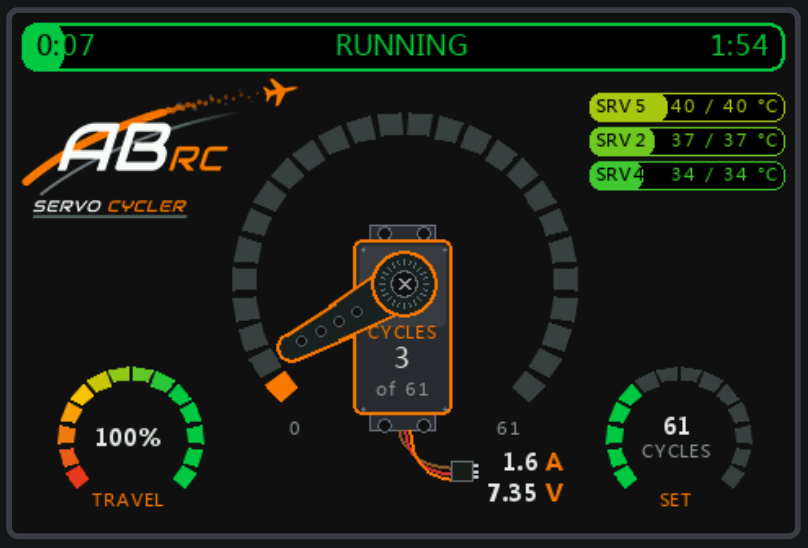
SERVO CYCLER

Servo tester for FrSky Ethos radios
Installation and user manual



Read the Disclaimer and Safety section before use. Servo Cycler is made to find poor servos, and a test can damage or destroy a servo. Use it only in a **separate test model** – never in a model that is flown or driven.

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github.com/adambednarik-ABRC/servo-cycler

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1. Disclaimer and safety

Disclaimer

Servo Cycler is provided free of charge, “as is”, without warranty of any kind, express or implied, including without limitation any warranty of fitness for a particular purpose, accuracy or reliability.

Servo Cycler is a test tool. Its purpose is to load servos continuously so that weak, worn or faulty servos show up. **A servo that fails during a test has done what the test is for.** Running a servo continuously, at full travel, for many cycles can overheat it, wear it out or destroy it. This applies above all to a servo that is already weak, under load, or blocked by a linkage or a mechanical stop.

By installing or using Servo Cycler you accept that:

- You use it entirely at your own risk.
- The author (ABRC) is not liable for any damage to servos, receivers, batteries, radios, models or other property, or for any injury, loss or consequential damage, however caused. This includes damage caused by incorrect setup, by use in a model that is used for flying or driving, by operating the tool while a model is in use, or by a fault in the script, the radio firmware or the hardware.
- You are responsible for setting up the radio correctly as described in this manual, and for checking that the setup is safe before every test.
- Test results such as temperature, current and voltage readings depend on your sensors and telemetry. They are given for information only and are not a guarantee that a servo is safe to fly.

Where the law does not allow some of these exclusions, they apply only as far as the law permits.

Why Servo Cycler must never be used in a model that flies

Servo Cycler moves every channel that uses it to its end points, again and again, for minutes at a time, regardless of the sticks. In a model that is flying or driving, that means total loss of control. Built-in protections:

- **Mixes:** it only moves channels whose mixes use the “Servo Cycler” source.
- **Lua source:** the source only works in a model where it is switched on under Model › Lua.
- **Widget:** a test can only run in the model that shows the Servo Cycler widget on its screen.
- **Switching models:** switching to another model stops a running test immediately.
- **Power-up:** if the start switch is already ON when the radio is switched on, or when you switch to the model, nothing starts until the switch is turned OFF and ON again.
- **Stopping:** turning the start switch OFF stops the test.

> These protections only work if the setup in this manual is followed. If the Servo Cycler mixes and widget are added to a flying model, the protections no longer separate testing from flying.

What you MUST do

- ✓ **Use a separate test model** used only for testing. Keep all Servo Cycler mixes, the widget and the Lua source in that model alone.
- ✓ **Use a spare switch that does nothing else** as the start switch – one you cannot flip by accident.
- ✓ **Set Slow up and Slow down to 0.5 s**, either in the Servo Cycler mixes or on the output channel (not both), in the test model only.
- ✓ **Check the free travel before the first test.** The servo must reach both end points without hitting a stop; a control surface or linkage must also move freely. Reduce the channel limits or the travel knob if needed. A servo pushing against a stop draws high current and overheats quickly.
- ✓ **Watch the test.** Stop it if a servo gets hot, buzzes, stalls or draws unusually high current.
- ✓ **Use a suitable power supply** that can run every servo under test continuously.
- ✓ **Make installed servos safe:** remove the propeller or rotor blades or disconnect the motor / ESC, secure the model, and keep people, hands and objects clear of moving parts.

What you MUST NOT do

- ✗ Never add the Servo Cycler source to a mix in a model that is flown or driven.
- ✗ Never put the Servo Cycler widget on a screen of a model that is flown or driven.
- ✗ Never start a test while a model is being operated – flying, taxiing, driving, or any time the motor can run.
- ✗ Never mix the Servo Cycler source into the throttle, ESC or motor channel, or into any channel that controls a motor, retracts, brakes or another function that could cause injury.
- ✗ Never use a start switch that is also used for flight functions such as flight modes, throttle cut or arming.
- ✗ Never leave a running test unattended.
- ✗ Never use Servo Cycler as proof that a servo is airworthy. A servo that passes the test can still fail later.

When something goes wrong

- **To stop:** turn the start switch OFF. The servos return to the centre at the slow rate, not instantly.
- **If the script stops with an error:** a red message appears at the bottom of the screen. A servo may then stay at its last position – switch off the receiver power if a servo stays under load.
- **If anything moves unexpectedly:** switch off the receiver power first, then check the setup.

2. What Servo Cycler does

Servo Cycler runs your servos from end to end for a set number of cycles (1 – 999) at a constant 30 cycles per minute, while the dashboard shows the progress and – if you have the sensors – servo temperatures, current and receiver voltage. One cycle is *centre › one end › other end › centre*.

The motion is made by the radio itself: the script only sets the next end point, and the **Slow up / Slow down** setting of the channel moves the servo there at an even rate. Screen refreshes can therefore never make a servo stutter, so any roughness you see or hear comes from the servo.

3. What you need

- A FrSky radio with Ethos and a colour touch screen (X20 family, X18 / Twin, X14, X10 / X12S).
- A receiver bound to a **separate test model**, with the servos to test and a suitable power supply.
- A spare switch for start / stop, and two pots: **pot 1** for travel and **pot 2** for the number of cycles (optional – fixed values can be used instead).
- Optional telemetry: up to 8 servo temperature sensors, a current sensor and an RX battery voltage sensor.

4. Installation

Download the latest release from **github.com/adambednarik-ABRC/servo-cycler/releases**.

A With the FrSky Suite (recommended)

1. Connect the radio to the computer and start the FrSky Suite.
2. Open **Lua library › Install from local .zip** and choose `ServoCycler-1.8.8-Suite.zip`.

B By hand

1. Unpack `ServoCycler-1.8.8-SD.zip`.
2. Connect the radio by USB (storage mode) and copy the `svcycler` folder from `scripts` into the `scripts` folder of the radio, so that you get `scripts/svcycler/main.lua`.
3. Restart the radio.

C Model picture (optional)

The package contains a model picture for each screen size in the folder `model-images`. Copy `svcycler.png` for your radio into `bitmaps/models/` (X20, X10 / X12S: SD card; X18 / Twin: internal storage, a copy on the SD card also works). Then choose it under **Model › Edit model › Picture**.

Folder	Radios	Size
X20_X20S_X20RS	X20, X20S, X20RS, X20 Pro	300 × 280
X18_X18S_Twin	X18, X18S, X18SE, Twin	180 × 166
X14	X14	240 × 224
X10_X12S	X10, X12S	180 × 160



Edit model › Picture



The test model on the home screen

5. Setting up the test model

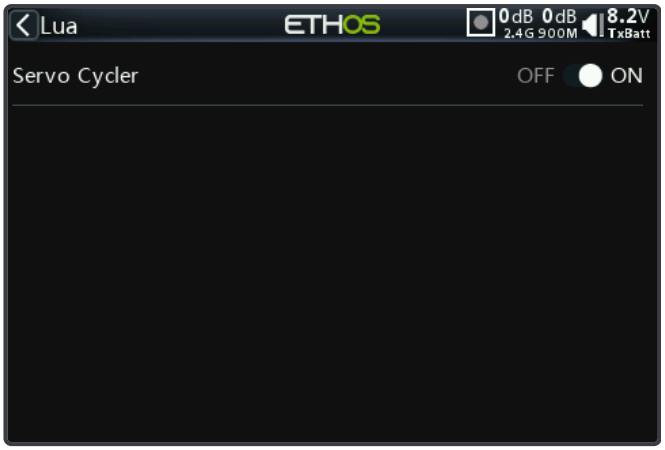
> Create a **new model** for testing (model type *Other* works well) and do all of the following in that model only.

1 Enable the Lua source

Open **Model › Lua** and switch **Servo Cycler** ON. Without this the source does nothing.



Model menu › Lua



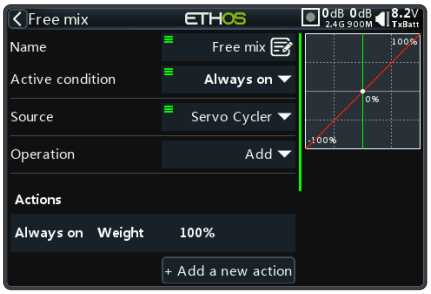
Servo Cycler ON

2 Add a mix for each servo channel

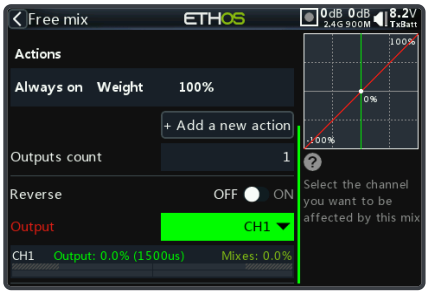
Open **Mixes**, tap **+** and add a **Free mix**:

- **Source:** Servo Cycler (category Lua) · **Active condition:** Always on · **Operation:** Add
- **Weight:** 100 % · **Outputs count:** 1 · **Output:** the channel of the servo (CH1, CH2 …)

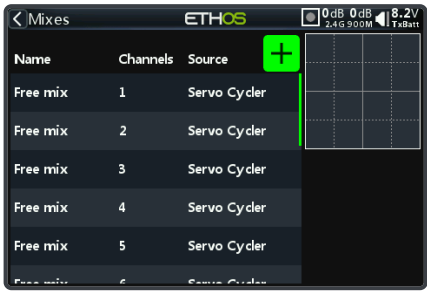
Repeat for every channel you want to test – up to 16 channels can run together.



Source Servo Cycler, weight 100 %



Output channel

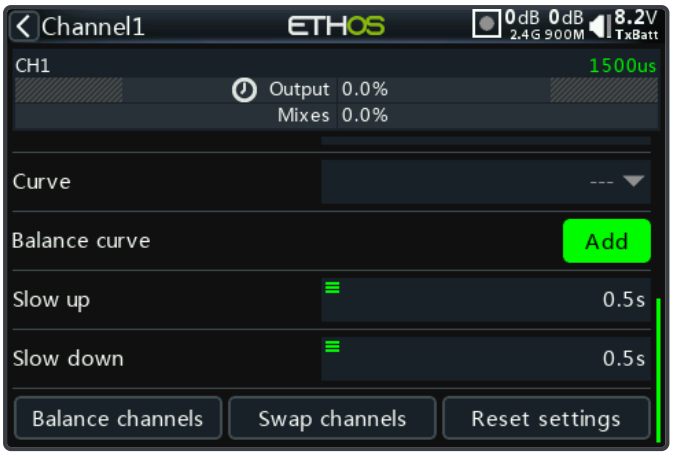


One Free mix per channel

3 Set Slow up and Slow down to 0.5 s

Open **Outputs**, select the channel and set **Slow up = 0.5 s** and **Slow down = 0.5 s**. Do this for every test channel. (You can set the slow in the mix instead – but **not in both places**, or the two add up and the servo no longer reaches its end points.)

This is what makes the motion smooth and gives full travel at exactly 30 cycles per minute: Ethos counts the slow time for 100 %, so 0.5 s is one full sweep from end to end in 1.0 s.

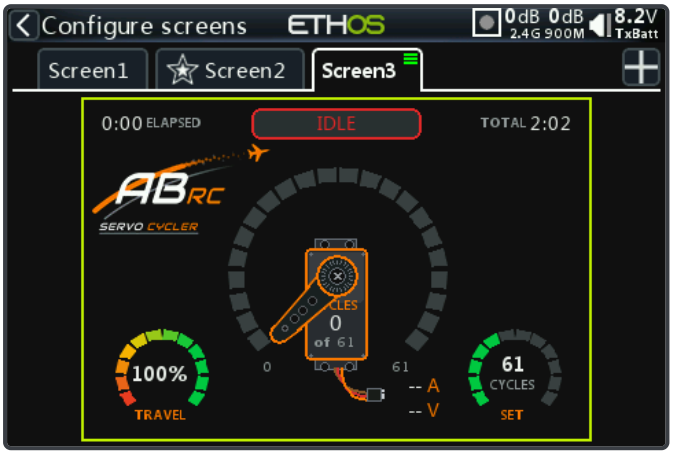


Outputs › channel › Slow up / Slow down 0.5 s

> Also check the channel **Min / Max** limits: at 100 % travel the servo must move freely to both ends without hitting a stop.

4 Add the widget

Open **Configure screens**, add a screen (**+**) with the **full-screen** layout (one widget) and choose the **Servo Cycler** widget. Set **Title** to OFF. The dashboard is designed for the whole screen.



A full-screen layout with the Servo Cycler widget

6. Configuring the widget

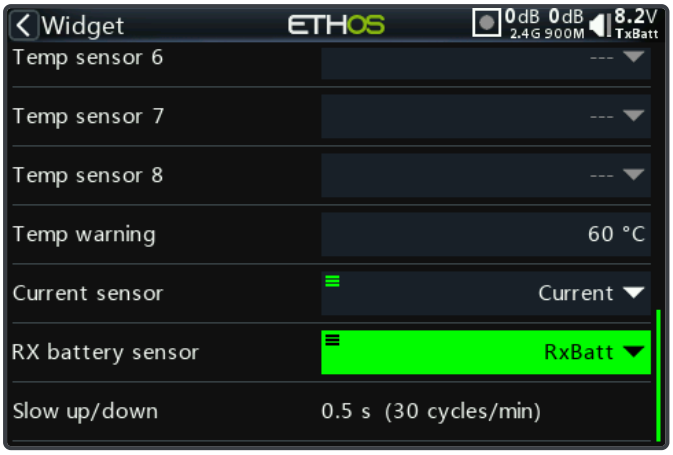
Long-press the widget (or open it from **Configure screens**) to reach its settings.



Switch, cycles and travel



Dials and temperature sensors



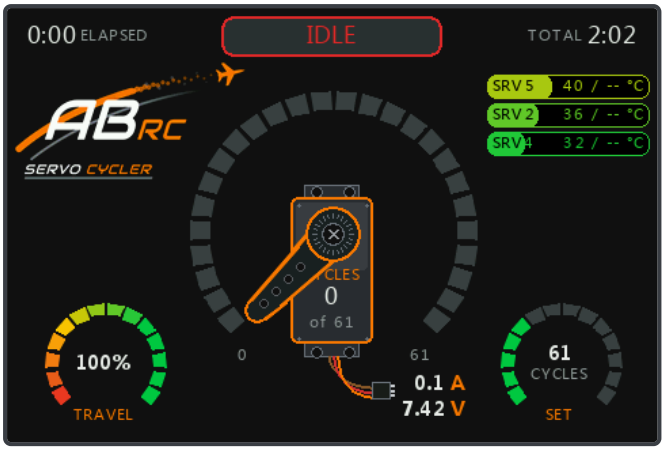
Warning limit, current, RX battery and slow reminder

Setting	What it does
Start switch	ON = start a test, OFF = stop. Use a spare switch that does nothing else. Without it the dashboard shows **NO SWITCH**.
Cycles knob (pot 2)	Sets the number of cycles, exponentially so small numbers are easy to set: 25 % = 17, 50 % = 126, 75 % = 422, full = 999. Frozen while a test runs. Use a knob or slider that is **calibrated** (Ethos › System › Hardware) so it reaches its full range. It picks an approximate number; for an exact number use the next line.
Cycles (no knob)	Exact number of cycles, used when **no** knob is selected above. Leave “Cycles knob” empty and type the number here (for example 61).
Travel knob (pot 1)	Travel from 0 % (centre) to 100 % (full travel).
Travel (no knob)	Exact travel, used when **no** knob is selected above. Leave “Travel knob” empty and type the value here.
Bottom dials	Travel left / Cycles right (above pot 1 and pot 2), or swapped.
Temp sensor 1 – 8	Servo temperature sensors from telemetry (optional). Pick the sensor from the list; it does not matter whether it is listed under *Telemetry* or *Telemetry active*. The 4 hottest sensors that are reporting are shown.
Temp warning	Temperature at which a temperature bar is fully red (default 60 °C).
Current sensor	Current sensor from telemetry (optional), shown as **A** at the servo plug. It shows the combined draw of all servos under test, so you can check it stays below the amp rating of the receiver.
RX battery sensor	Receiver voltage sensor from telemetry (optional), shown as **V** at the servo plug. Watch it during a long test so the battery is not deep-discharged.
Slow up/down	Reminder of the slow setting the test model needs: 0.5 s (30 cycles/min).

7. Running a test

1. Check that the servos can move freely and that the model is safe (see section 1).
2. Set the number of cycles with **pot 2** and the travel with **pot 1**. The dashboard shows the total test time on the right.
3. Switch the start switch **ON**. The top bar turns green and fills up from left to right as the test runs.
4. Switch it **OFF** at any time to stop. The servos return to the centre and the result stays on the screen (**STOPPED**).
5. When all cycles are done the dashboard shows **DONE** and the servos stop at the centre. Switch OFF and ON again to run another test.

> If the start switch is already ON when you switch the radio on or change to the test model, nothing starts until you switch it OFF and ON again.



IDLE – ready, total time shown



RUNNING – the bar fills with the progress



DONE – all cycles completed

8. The dashboard explained



Area	Meaning
Top bar (running)	Elapsed time on the left, **RUNNING** in the middle, time left on the right. The bar fills green from left to right with the progress.
Top bar (not running)	Status box: **IDLE**, **STOPPED**, **DONE** or **NO SWITCH**, with the elapsed time on the left and the total time (or time left after a stop) on the right.
Top left	ABRC Servo Cycler logo.
Top right	Temperature bars of the 4 hottest sensors: current / highest temperature of the test. The bar fills and turns from green to red as the temperature approaches the warning limit. Only sensors with live telemetry are shown.
Centre dial	Progress of the test from 0 to the set number of cycles. The servo arm is the pointer; the case shows the cycles done “of” the target.
Servo plug	Current (**A**) and receiver voltage (**V**), if the sensors are set.
Bottom left	**TRAVEL** dial (pot 1): red at low travel, green from 66 %.
Bottom right	**CYCLES SET** dial (pot 2): number of cycles of the next test; green up to 66 % of the knob, then towards red for long tests.
Red line at the bottom	A script error message. It disappears after 15 s. Please report it (see section 10).

While a test runs the screen refreshes about once a second, right after each turn of the servos. The servo motion itself is made by the radio and is not affected by the screen.

9. Troubleshooting

Problem	Check
Servos do not move	Model › Lua: Servo Cycler ON? A Free mix with source Servo Cycler on that channel? Start switch ON (after OFF)? Travel knob above 0 %?
NO SWITCH on the dashboard	Set the start switch in the widget configuration.
Nothing starts with the switch ON	By design after power-up or a model change: switch OFF and ON again. After DONE: switch OFF and ON to run again.
Servo does not reach its end points	Slow longer than 0.5 s, or slow set both in the mix and on the channel. Channel Min / Max limits too small. Travel knob below 100 %.
Servo pauses at the end points	Slow shorter than 0.5 s. Set exactly 0.5 s up and down. (At travel below 100 % a short pause at the ends is normal.)
Movement is rough or jerky	Slow not set (the servo jumps from end to end), or the servo itself – which is what the test is for.
Cycles stuck above a number (e.g. 123)	The cycles input must be a knob or slider, and it must be calibrated so it reads its full range (Ethos › System › Hardware). A stick, or a pot that rests mid-position, cannot reach the low numbers. The knob only picks an approximate number. For an exact number leave “Cycles knob” empty and type the number in “Cycles (no knob)”.
Not sure what to select for the sensors	All sensor fields are optional and only show readings; they do not change how the servos move. Pick the sensor from either the *Telemetry* or the *Telemetry active* list, or leave the field empty.
What are A and V for?	A is the current drawn by all servos together: check that it stays below the amp rating of your receiver. V is the receiver battery voltage: watch it so the battery is not deep-discharged during a long test.
No temperature bars	Sensors not set, or no telemetry from them at the moment. Only reporting sensors are shown.
A / V show “--”	Sensor set but no telemetry received.
Red error line	Note the message and report it with your radio model and Ethos version.

10. License and support

Servo Cycler is free software under the **MIT license**, © 2026 Adam Bednarik (ABRC). The ABRC name and the ABRC / Servo Cycler logos and images are not covered by the license.

Questions, problems and ideas: open an issue at **github.com/adambednarik-ABRC/servo-cycler/issues**. Please include your radio model, Ethos version and, if possible, a screenshot.

Servo Cycler is an independent project and is not affiliated with FrSky. FrSky and Ethos are names of their respective owners.

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